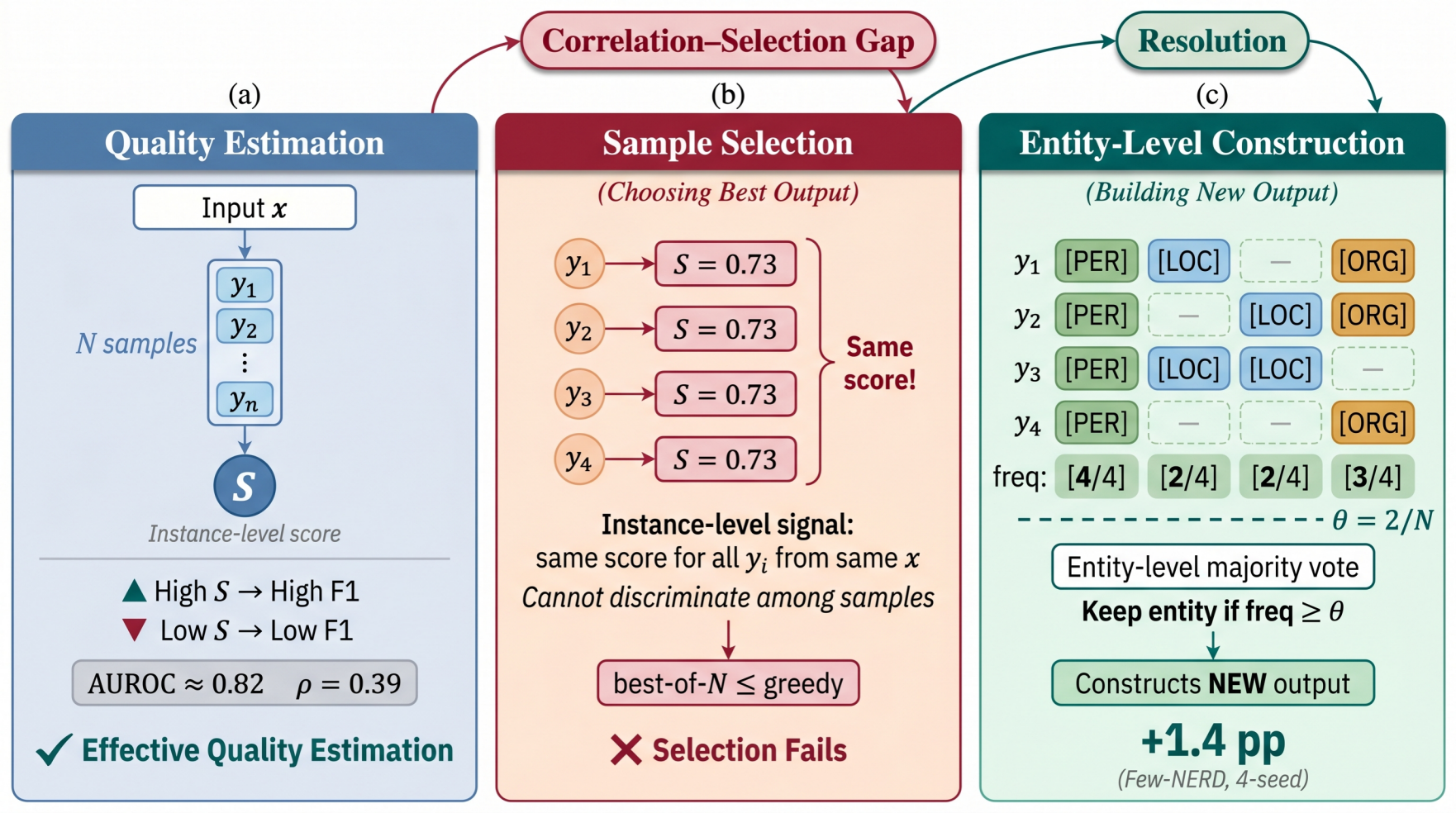

Correlation–Selection Gap
Resolution
(a)
Quality Estimation
Input $x$
$N$ samples
$y_1$
$y_2$
$y_n$
$S$
Instance-level score
▲ High $S$ → High F1
▼ Low $S$ → Low F1
AUROC ≈ 0.82 $\rho = 0.39$
✓ Effective Quality Estimation
(b)
Sample Selection
(Choosing Best Output)
$y_1$ $S = 0.73$
$y_2$ $S = 0.73$
$y_3$ $S = 0.73$
$y_4$ $S = 0.73$
Same score!
Instance-level signal:
same score for all $y_i$ from same $x$
Cannot discriminate among samples
best-of-$N$ ≤ greedy
✗ Selection Fails
(c)
Entity-Level Construction
(Building New Output)
$y_1$ [PER] [LOC] — [ORG]
$y_2$ [PER] — [LOC] [ORG]
$y_3$ [PER] [LOC] [LOC] —
$y_4$ [PER] — — [ORG]
freq: [4/4] [2/4] [2/4] [3/4]
$\theta = 2/N$
Entity-level majority vote
Keep entity if freq ≥ $\theta$
Constructs NEW output
+1.4 pp
(Few-NERD, 4-seed)